

Assignment 06 — Bioinformatics Pipeline

BIOS 30318/60318: Introduction to Biocomputing, Fall 2026

Week 8 | Lectures L13–L14 | Tutorial T07

Overview

Build and execute a bioinformatics pipeline using sequence alignment (**muscle**) and profile hidden Markov models (**hmm**) to search for a gene family across multiple bacterial proteomes.

Points: 50 | Difficulty: | Midterm Assignment

Background

You will use **muscle** for multiple sequence alignment and **hmm** (**hmm**build + **hmm**search) for profile-based sequence searching. The reference sequences and proteomes are available in the **bioinformaticsProject/** directory.

Exercises

1 (8 pts)

- a) Select a set of reference sequences from the **ref_sequences/** directory. Use **muscle** to create a multiple sequence alignment. Show the command and briefly describe the output format.
- b) Use **hmm**build to build an HMM profile from your alignment. Report the command and the name of your profile.

2 (12 pts)

- a) Use **hmm**search to search your HMM profile against **one** proteome from the **proteomes/** directory. Save the output with **--tblout**.
- b) Examine the output. How many significant hits (E-value < 0.001) did you find? Show the **grep** or **awk** command used to filter.
- c) Explain in 2–3 sentences what the E-value represents and why we use a threshold.

3 (13 pts)

Write a bash script called `run_pipeline.sh` that:

1. Takes a reference FASTA file and a directory of proteome files as arguments
2. Builds the alignment and HMM profile
3. Searches **every** proteome in the directory
4. Outputs a CSV file `pipeline_results.csv` with columns: `proteome,num_hits`

Include error checking (e.g., verify files exist). Test on at least 5 proteomes.

4 (9 pts)

- a) Run your pipeline on all proteomes in the `proteomes/` directory.
- b) Which proteome had the most hits? Which had the fewest (or zero)?
- c) What biological question could the variation in hit counts answer?

5 (8 pts)

- a) Describe the difference between **pairwise alignment** and **profile-based searching**. When would you use each?
 - b) Why is building an HMM profile from multiple sequences more sensitive than using a single sequence as a BLAST query?
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Submission

Submit `run_pipeline.sh`, `pipeline_results.csv`, and `assignment06_NETID.txt` with commands, outputs, and explanations.